

Culminating Activity

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Abstract

Provide a concise summary of the study including purpose, methods, results, and conclusions (150–250 words).

Chapter 1: Introduction

Background of the Study

Global sales prediction plays an important role in helping businesses make informed decisions related to marketing strategies, inventory management, budgeting, and future business planning. Many organizations rely on predictive models such as Multiple Linear Regression to analyze trends and forecast sales performance using historical and market-related data. However, one of the common challenges in predictive analytics is the presence of missing data within datasets. Missing values may occur due to incomplete records, human error, system failures, or inconsistent data collection processes. When these missing values are not properly handled, they can negatively affect the accuracy, consistency, and reliability of sales prediction models, leading to poor decision-making and inaccurate forecasts.

This research focuses on examining the effect of missing value handling on improving global sales prediction using Multiple Linear Regression. Specifically, it aims to understand why missing data reduces model performance and how different data cleaning techniques can enhance predictive accuracy. It also highlights the significance of proper data preparation in machine learning and statistical analysis. By identifying effective preprocessing methods, the research can help businesses, researchers, and data analysts develop more reliable prediction models and improve the quality of data-driven decisions in global sales forecasting.

Statement of the Problem

1. What is the effect of missing value handling on improving global sales prediction using Multiple Linear Regression?
2. Why does missing data reduce the accuracy and reliability of sales prediction models?
3. How does applying data cleaning techniques improve the performance of the Multiple Linear Regression model?

Objectives of the Study

General Objective

- To investigate the effect of missing value handling on improving global sales prediction using Multiple Linear Regression.

Specific Objectives

- To determine how missing data affects the accuracy and reliability of sales prediction models.
- To implement and assess data imputation strategies to prevent the loss of valuable data caused by deleting incomplete rows.
- To compare the performance metrics between models using data with deleted missing rows and models using imputed data.

Significance of the Study

Explain who benefits and how.

Scope and Limitations

Define the boundaries of the study.

Chapter 2: Review of Related Literature

Related Studies

Summarize previous research.

Related Literature

Discuss theories and concepts.

Conceptual Framework

(Insert diagram if needed)

Chapter 3: Methodology

Research Design

Describe the design (e.g., descriptive, experimental).

Data Collection

The dataset utilized in this study was collected from Kaggle, <https://www.kaggle.com/code/yonatanrabinovich/video-games-sales-regression-techniques/input>.

```
# Load dataset
data <- read.csv("../Video_Games_Sales_as_at_22_Dec_2016.csv", sep=",", na.strings=c("", " ", "NA", "N/A"))
head(data)
```

##	Name	Platform	Year_of_Release	Genre	Publisher
## 1	Wii Sports	Wii	2006	Sports	Nintendo
## 2	Super Mario Bros.	NES	1985	Platform	Nintendo
## 3	Mario Kart Wii	Wii	2008	Racing	Nintendo
## 4	Wii Sports Resort	Wii	2009	Sports	Nintendo
## 5	Pokemon Red/Pokemon Blue	GB	1996	Role-Playing	Nintendo
## 6	Tetris	GB	1989	Puzzle	Nintendo

##	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales	Critic_Score	Critic_Count
## 1	41.36	28.96	3.77	8.45	82.53	76	51
## 2	29.08	3.58	6.81	0.77	40.24	NA	NA
## 3	15.68	12.76	3.79	3.29	35.52	82	73
## 4	15.61	10.93	3.28	2.95	32.77	80	73
## 5	11.27	8.89	10.22	1.00	31.37	NA	NA
## 6	23.20	2.26	4.22	0.58	30.26	NA	NA

##	User_Score	User_Count	Developer	Rating
## 1	8	322	Nintendo	E
## 2	<NA>	NA	<NA>	<NA>
## 3	8.3	709	Nintendo	E
## 4	8	192	Nintendo	E
## 5	<NA>	NA	<NA>	<NA>
## 6	<NA>	NA	<NA>	<NA>

Variables Description

Explain dependent and independent variables.

Data Analysis Techniques

- Descriptive Statistics
- Inferential Statistics
- Machine Learning Models (if applicable)

Chapter 4: Results and Discussion

Descriptive Statistics

Visualization

```
library(ggplot2)
#ggplot(data, aes(x=Variable1, y=Variable2)) +
#  geom_point() +
#  theme_minimal()
```

Model

```
#model <- lm(total_sales~., numeric_data)
#summary(model)
```

Interpretation

Discuss findings in paragraph form.

Chapter 5: Conclusion and Recommendations

Conclusion

Summarize key findings.

Recommendations

Provide practical suggestions

References

Rabinovich, Y. (2021). *Video games sales regression techniques*. Kaggle. <https://www.kaggle.com/code/yonatanrabinovich/video-games-sales-regression-techniques/notebook>

The Investopedia Team. (2025). *Predictive Modeling: Techniques, Uses, and Key Takeaways*. <https://www.investopedia.com/terms/p/predictive-modeling.asp>

Appendices

Appendix A: Code